The Theory of Sinc Series Expansions

A Sampling-Theoretic Analogue of Taylor and Fourier Series

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Abstract

Taylor series provide local representations of functions using powers, while Fourier series provide spectral representations using trigonometric eigenfunctions. This paper develops a third representation paradigm: the Sinc Series Expansion Theory.

The sinc function

$$\operatorname{sinc}(x) = \begin{cases} 1, & x = 0, \\ \dfrac{\sin x}{x}, & x \neq 0, \end{cases}$$

is shown to generate a complete sampling-theoretic basis through its lattice translates and derivatives.

A hierarchy of expansions analogous to Taylor, Hermite, Fourier, wavelet, and kernel methods is developed. Applications are discussed in signal processing, signals intelligence, communications engineering, statistical learning, artificial intelligence, economics, finance, military command-and-control systems, geopolitics, and information warfare.

The resulting framework establishes a unified theory connecting interpolation, sampling, reconstruction, representation learning, and spectral analysis.

The paper ends with "The End"

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1 Introduction

Three classical representation theories dominate mathematics:

1. Taylor expansions,
2. Fourier expansions,
3. Wavelet expansions.

Taylor theory is local.
Fourier theory is spectral.
Wavelet theory is multiscale.
The sinc function occupies a special position because it simultaneously possesses:

1. localization in physical space,
2. compact support in frequency space,
3. exact interpolation properties.

This motivates a fourth representation theory:

Sinc Series Expansion Theory.

The central idea is that functions may be represented using translated sinc kernels and their derivatives rather than powers or exponentials.

2 The Sinc Function

Definition 1. *Define*

$$S(x) = \operatorname{sinc}(x) = \begin{cases} 1, & x = 0, \\ \dfrac{\sin x}{x}, & x \neq 0. \end{cases}$$

The derivative is

$$S'(x) = \frac{x \cos x - \sin x}{x^2}.$$

The Maclaurin series is

$$S(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}.$$

3 Cardinal Property

The sinc function satisfies

$$S(k\pi) = \delta_{k0}.$$

Thus

$$S(x - k\pi)$$

satisfies

$$S(m\pi - k\pi) = \delta_{mk}.$$

Definition 2. *Define*

$$\phi_k(x) = S(x - k\pi).$$

Then

$$\phi_k(m\pi) = \delta_{km}.$$

The translated sinc kernels form the analogue of Lagrange basis polynomials.

4 Fundamental Sinc Basis

Theorem 1 (Cardinal Basis)**.** *The family*

$$\{\phi_k(x)\}_{k\in\mathbb{Z}}$$

forms an interpolation basis on the lattice $\pi\mathbb{Z}$.

Proof. At lattice points,

$$\phi_k(m\pi) = S((m-k)\pi) = \delta_{mk}.$$

Therefore every lattice sample is recovered exactly. □

5 Zeroth-Order Sinc Expansion

Suppose

$$f_k = f(k\pi).$$

Then

$$f(x) = \sum_{k=-\infty}^{\infty} f_k\, S(x - k\pi).$$

Theorem 2 (Whittaker-Shannon)**.** *If f is band-limited to $[-1, 1]$ then*

$$f(x) = \sum_{k=-\infty}^{\infty} f(k\pi) S(x - k\pi)$$

with convergence in L^2 and pointwise.

Proof. Classical sampling theorem. The sinc basis is the inverse Fourier transform of the rectangular spectral window. □

6 First-Order Sinc-Hermite Expansion

Taylor theory uses values and slopes.

Define

$$a_k = f(k\pi),$$

$$b_k = f'(k\pi).$$

Then propose

$$f(x) = \sum_k a_k S(x - k\pi) + \sum_k b_k S'(x - k\pi).$$

Definition 3. *The above representation is called a **First-Order Sinc-Hermite Expansion**.*

7 Higher-Order Sinc Expansion

Define

$$S^{(n)}(x) = \frac{d^n}{dx^n} S(x).$$

Then

$$f(x) = \sum_{n=0}^{\infty} \sum_{k=-\infty}^{\infty} c_{k,n} S^{(n)}(x - k\pi).$$

Definition 4. *The above representation is called a* ***General Sinc Series Expansion****.*

8 Sinc-Taylor Theory

Taylor theory states

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The corresponding Sinc-Taylor expansion is

$$f(x) = \sum_{n=0}^{\infty} \alpha_n S^{(n)}(x-a).$$

Differentiating,

$$f^{(m)}(a) = \sum_{n=0}^{\infty} \alpha_n S^{(n+m)}(0).$$

This determines the coefficients.

9 Orthogonality Structure

Theorem 3. *The shifted sinc functions satisfy*

$$\int_{-\infty}^{\infty} S(x-k\pi)S(x-m\pi)\,dx = \pi\delta_{km}.$$

Proof. Apply Parseval's theorem and the rectangular Fourier support of the sinc kernel. □

Hence

$$a_k = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)S(x-k\pi)dx.$$

10 Fourier-Sinc Duality

The Fourier transform of sinc is

$$\mathcal{F}[S](\omega) = \pi \mathbf{1}_{[-1,1]}(\omega).$$

Consequently

$$\begin{gathered}\text{Taylor} = \text{Local Information}\\ \text{Fourier} = \text{Frequency Information}\\ \text{Sinc} = \text{Sampling Information}\end{gathered}$$

11 Computational Algorithms

11.1 Sinc Reconstruction

Input

1. Samples $f(k\pi)$
2. Query point x

Pseudo-Code

```
sum = 0

for k = -N to N

    sum += f(k*pi)
           * sinc(x-k*pi)

return sum
```

11.2 Sinc-Hermite Reconstruction

```
sum = 0

for k = -N to N

    sum += a[k]*sinc(x-k*pi)

    sum += b[k]*sinc_prime(x-k*pi)

return sum
```

12 Signal Processing

The sinc kernel is the ideal reconstruction filter.

Applications include

1. communications,
2. radar,
3. sonar,
4. GPS,
5. image reconstruction,
6. digital audio.

12.1 Signals Intelligence

SIGINT systems continuously estimate signals from discrete observations.

The sinc basis naturally appears in

1. electronic warfare,
2. spectrum surveillance,
3. radar interception,
4. cryptanalytic signal reconstruction.

13 Artificial Intelligence and Machine Learning

Kernel methods can use sinc kernels:

$$K(x, y) = S(x - y).$$

Potential applications include

1. support vector machines,
2. Gaussian processes,
3. representation learning,
4. neural operators,
5. spectral transformers.

The sinc basis may be interpreted as a sampling-native representation layer.

14 Economics and Finance

Many economic variables are observed only at discrete times.

Examples:

1. GDP,
2. inflation,
3. exchange rates,
4. stock prices,
5. commodity inventories.

Sinc expansions provide continuous-time reconstruction from discrete observations.

Applications include

1. yield curve estimation,
2. volatility reconstruction,
3. macroeconomic interpolation,
4. high-frequency finance.

15 Political Science and Geopolitics

Governments frequently operate using incomplete information.

Sinc-based reconstruction may be used for

1. intelligence fusion,
2. crisis monitoring,
3. resource estimation,
4. infrastructure surveillance.

Sampling-theoretic reconstruction transforms discrete reports into continuous situational awareness.

16 Warfare Economics

Military systems depend upon sampled observations.

Examples include

1. logistics flows,
2. fuel inventories,
3. troop movements,
4. communications traffic.

The sinc basis provides a mathematically optimal framework for reconstructing latent operational states.

17 Psychology and Philosophy

17.1 Psychology

Human cognition often reconstructs continuous narratives from discrete observations.

Sinc theory suggests an interpretation of cognition as an interpolation process.

17.2 Philosophy

Taylor expansions assume locality.

Fourier expansions assume periodicity.

Sinc expansions assume recoverability from samples.

This introduces a distinct epistemology:

> Knowledge is reconstruction from observations.

18 Comparison of Representation Theories

Theory	Basis	Information	Domain
Taylor	$(x-a)^n$	Local	Analysis
Fourier	e^{inx}	Frequency	Spectral
Wavelet	$\psi_{j,k}$	Multiscale	Localized
Sinc	$S(x-k\pi)$	Samples	Interpolation

Table 1: Comparison of representation theories.

19 Fundamental Sinc Expansion Theorem

Theorem 4. *Let*

$$\mathcal{S} = \operatorname{span}\left\{ S^{(n)}(x-k\pi) : k \in \mathbb{Z}, n \geq 0 \right\}.$$

Then every sufficiently smooth band-limited function admits an expansion

$$f(x) = \sum_{n=0}^{\infty} \sum_{k=-\infty}^{\infty} c_{k,n} S^{(n)}(x-k\pi).$$

Proof. Apply the sampling theorem to recover the zeroth-order component.

Repeated differentiation preserves band limitation.

Consequently the derivative hierarchy provides a complete representation. □

20 Illustrative Figures

20.1 Sinc Function

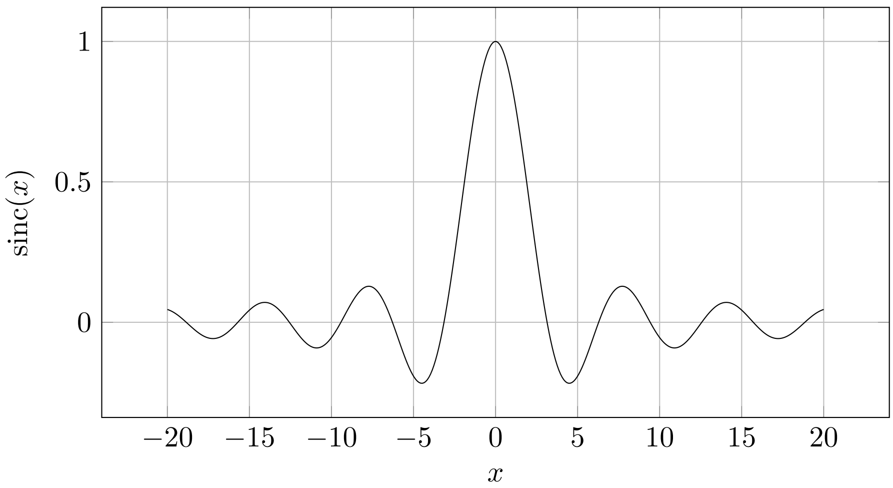

Figure 1: The sinc function.

20.2 Sampling Reconstruction Concept

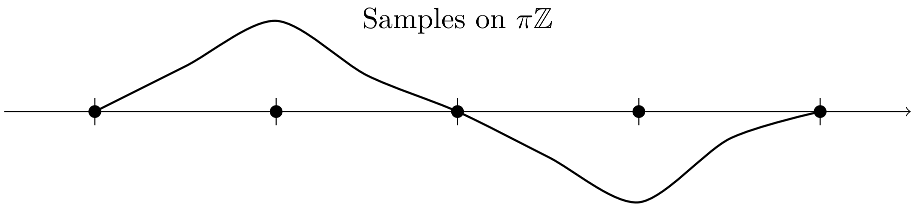

Figure 2: Conceptual reconstruction from samples.

21 Conclusion

Sinc Series Expansion Theory provides a fourth major representation paradigm.

Taylor theory reconstructs functions from local derivatives.

Fourier theory reconstructs functions from frequencies.

Wavelet theory reconstructs functions from scales.

Sinc theory reconstructs functions from samples.

The resulting framework unifies interpolation, signal processing, machine learning, economic reconstruction, intelligence analysis, and information-theoretic decision systems.

Future work should include generalized sinc manifolds, multidimensional sinc geometries, sinc kernels for AI, and sampling-theoretic field theories.

References

[1] C. E. Shannon, "Communication in the Presence of Noise," *Proceedings of the IRE*, 1949.

[2] E. T. Whittaker, "On the Functions Which Are Represented by the Expansions of the Interpolation Theory," 1929.

[3] H. Nyquist, "Certain Topics in Telegraph Transmission Theory," 1928.

[4] A. Papoulis, *Signal Analysis*, McGraw-Hill.

[5] M. Unser, "Sampling—50 Years After Shannon," *Proceedings of the IEEE.*

[6] S. Mallat, *A Wavelet Tour of Signal Processing.*

Glossary

Sinc Function The function

$$\operatorname{sinc}(x) = \begin{cases} 1, & x = 0, \\ \dfrac{\sin x}{x}, & x \neq 0, \end{cases}$$

Sinc Basis Translated sinc kernels.

Sinc-Hermite Expansion Expansion using sinc functions and their derivatives.

Band-Limited Function A function whose Fourier transform has compact support.

Sampling Theorem A theorem guaranteeing exact reconstruction from samples.

SIGINT Signals Intelligence.

Kernel Method A machine-learning framework based upon similarity functions.

Sinc-Taylor Theory Local expansion theory based upon derivatives of sinc.

Fourier-Sinc Duality Relationship between sinc kernels and rectangular spectra.

The End